ROBOTICS

ENGR – UH 4330

Fall, 2025

Experiment: **Lab 7 - Jacobian**

Due date: **12 / NOV / 2025**

Section Number: 001

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1. Introduction

In robotic systems, the **Jacobian** defines the relationship between **joint velocities** and the **end-effector's linear and angular velocities**. Mathematically, this relationship is given by:

$$v_e = J(q)\dot{q}$$

where $v_e = [\dot{p}_e^T \quad \omega_e^T]$ is the end-effector velocity vector, and $J(q)$ is the Jacobian matrix that depends on the joint configuration q

Configurations at which $J(q)$ loses rank are called **kinematic singularities**. These configurations are crucial to study for the following reasons:

1. **Reduced mobility:** At singularities, the manipulator loses degrees of freedom, restricting its motion.
2. **Infinite inverse kinematics solutions:** Multiple joint configurations may result in the same end-effector pose.
3. **Amplified joint velocities:** Near singularities, small end-effector motions may require extremely large joint velocities.
4. **Boundary singularities:** These occur when the manipulator is fully stretched or folded.
5. **Internal singularities:** Occur due to alignment of axes within the workspace and are more problematic since they may occur along the manipulator's path.

This lab aims to visualize, identify, and interpret singularities both **mathematically** (through Jacobian analysis) and **physically** (via simulation in Simscape).

2. Lab Objectives

1. Derive and analyze the Jacobian for a 3-link planar manipulator.
2. Identify and classify the manipulator’s singularities.
3. Extend the study to an **anthropomorphic arm** to explore **shoulder** and **boundary** singularities.
4. Examine the effect of singularity proximity on joint velocities during motion.

3. Results

3.1. Three-Link Planar Manipulator

3.1.1 Symbolic Derivation

Using MATLAB's symbolic toolbox, the transformation matrices for each joint were derived and used to compute the **geometric Jacobian**.

Figure 1: Transformation chain for 3-link planar manipulator

$$\begin{pmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & \frac{2\cos(\theta_1)}{5} \\ \sin(\theta_1) & \cos(\theta_1) & 0 & \frac{2\sin(\theta_1)}{5} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos(\theta_2) & -\sin(\theta_2) & 0 & \frac{\cos(\theta_2)}{5} \\ \sin(\theta_2) & \cos(\theta_2) & 0 & \frac{\sin(\theta_2)}{5} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \cos(\theta_3) & -\sin(\theta_3) & 0 & \frac{\cos(\theta_3)}{10} \\ \sin(\theta_3) & \cos(\theta_3) & 0 & \frac{\sin(\theta_3)}{10} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

(a) A[0,1] (b) A[1,2] (c) A[0,3]

3.1.2 Determinant of the Jacobian

The determinant of the Jacobian was obtained symbolically as a function of the joint angles.

Figure 2.1: Geometric and Analytical Jacobian as function of joint angles (θ_1, θ_2, θ_3)

$$\begin{pmatrix} -\sigma_2 - \frac{\sin(\theta_1+\theta_2)}{5} - \frac{2\sin(\theta_1)}{5} & -\sigma_2 - \frac{\sin(\theta_1+\theta_2)}{5} & -\sigma_2 \\ \sigma_1 + \frac{\cos(\theta_1+\theta_2)}{5} + \frac{2\cos(\theta_1)}{5} & \sigma_1 + \frac{\cos(\theta_1+\theta_2)}{5} & \sigma_1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

where

$$\sigma_1 = \frac{\cos(\theta_1+\theta_2+\theta_3)}{10}$$

$$\sigma_2 = \frac{\sin(\theta_1+\theta_2+\theta_3)}{10}$$

(a). Geometric Jacobian

$$\begin{pmatrix} -\sigma_2 - \frac{\sin(\theta_1+\theta_2)}{5} - \frac{2\sin(\theta_1)}{5} & -\sigma_2 - \frac{\sin(\theta_1+\theta_2)}{5} & -\sigma_2 \\ \sigma_1 + \frac{\cos(\theta_1+\theta_2)}{5} + \frac{2\cos(\theta_1)}{5} & \sigma_1 + \frac{\cos(\theta_1+\theta_2)}{5} & \sigma_1 \\ 1 & 1 & 1 \end{pmatrix}$$

where

$$\sigma_1 = \frac{\cos(\theta_1+\theta_2+\theta_3)}{10}$$

$$\sigma_2 = \frac{\sin(\theta_1+\theta_2+\theta_3)}{10}$$

(b). Analytical Jacobian

Figure 2.2: Determinant of Jacobian as function of joint angles (θ_1, θ_2, θ_3)

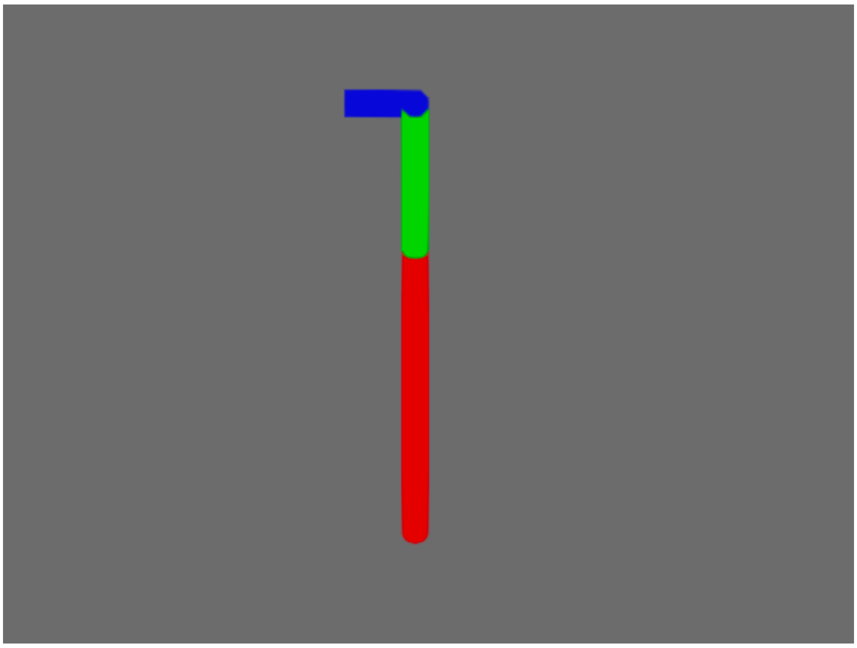
$$\frac{2\sin(\theta_2)}{25}$$

3.1.3 Singular Configurations

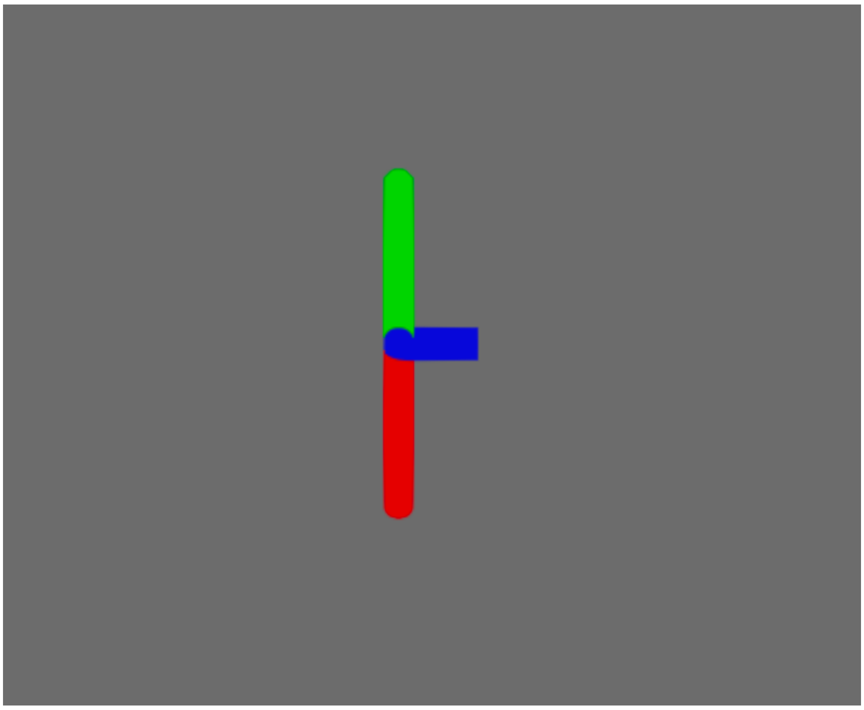
The singularities were identified where det(J) = 0.

$$\theta_2 = \pi k$$

Figure 3: Manipulator configurations at singular positions



(a). $\theta_2 = 0$



(b). $\theta_2 = \pi$

3.1.4 Classification

The found singularities are **boundary singularities**, occurring when the arm is either fully extended or folded back onto itself.

3.2. Anthropomorphic Arm Analysis

3.2.1 Symbolic Jacobian Derivation

Using the symbolic toolbox, the full Jacobian for the 3-DOF anthropomorphic arm was derived.

Figure 4.1: Jacobian structure for anthropomorphic arm

$$\begin{pmatrix} -\frac{\sin(\theta_1)\,\sigma_2}{2} & -\cos(\theta_1)\,\sigma_1 & -\frac{\sin(\theta_2+\theta_3)\cos(\theta_1)}{2} \\ \frac{\cos(\theta_1)\,\sigma_2}{2} & -\sin(\theta_1)\,\sigma_1 & -\frac{\sin(\theta_2+\theta_3)\sin(\theta_1)}{2} \\ 0 & \sigma_3 + \frac{\cos(\theta_2)}{2} & \sigma_3 \\ 0 & \sin(\theta_1) & \sin(\theta_1) \\ 0 & -\cos(\theta_1) & -\cos(\theta_1) \\ 1 & 0 & 0 \end{pmatrix}$$

where

$$\sigma_1 = \frac{\sin(\theta_2+\theta_3)}{2} + \frac{\sin(\theta_2)}{2}$$

$$\sigma_2 = \cos(\theta_2+\theta_3) + \cos(\theta_2)$$

$$\sigma_3 = \frac{\cos(\theta_2+\theta_3)}{2}$$

Figure 4.2: Reduced Jacobian structure

$$\begin{pmatrix} -\frac{\sin(\theta_1)\,\sigma_2}{2} & -\cos(\theta_1)\,\sigma_1 & -\frac{\sin(\theta_2+\theta_3)\cos(\theta_1)}{2} \\ \frac{\cos(\theta_1)\,\sigma_2}{2} & -\sin(\theta_1)\,\sigma_1 & -\frac{\sin(\theta_2+\theta_3)\sin(\theta_1)}{2} \\ 0 & \frac{\cos(\theta_2+\theta_3)}{2} + \frac{\cos(\theta_2)}{2} & \frac{\cos(\theta_2+\theta_3)}{2} \end{pmatrix}$$

where

$$\sigma_1 = \frac{\sin(\theta_2+\theta_3)}{2} + \frac{\sin(\theta_2)}{2}$$

$$\sigma_2 = \cos(\theta_2+\theta_3) + \cos(\theta_2)$$

Figure 4.3: Determinant of Jacobian as function of joint angles (θ_1, θ_2, θ_3)

$$-\frac{(\cos(\theta_3)+1)\ (\sin(\theta_2+\theta_3)-\sin(\theta_2))}{8}$$

3.2.2 Types of Singularities

Two primary singularities were identified:

- **Boundary singularity:** Arm fully stretched (end of reachable workspace)

$$\begin{pmatrix} \pi\ (2\,k+1) \\ z \\ 2\,\pi\,k \end{pmatrix}$$

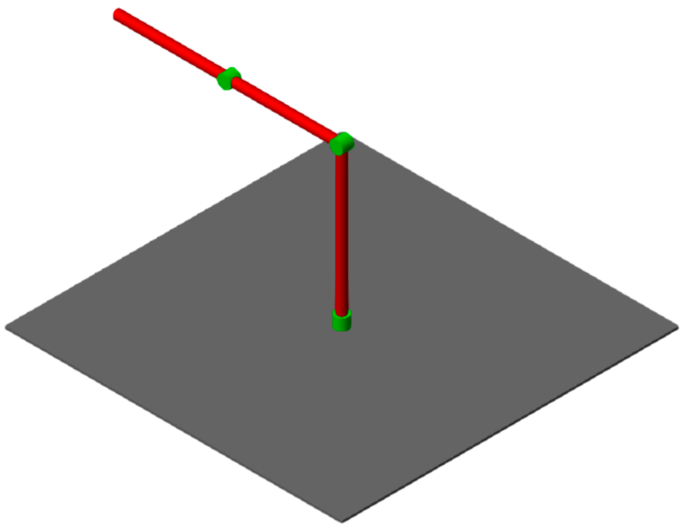
- **Shoulder singularity:** Axis alignment between the first two joints

$$\begin{pmatrix} z \\ \frac{\pi}{2}-\frac{z}{2}+\pi\,l \\ z \end{pmatrix}$$

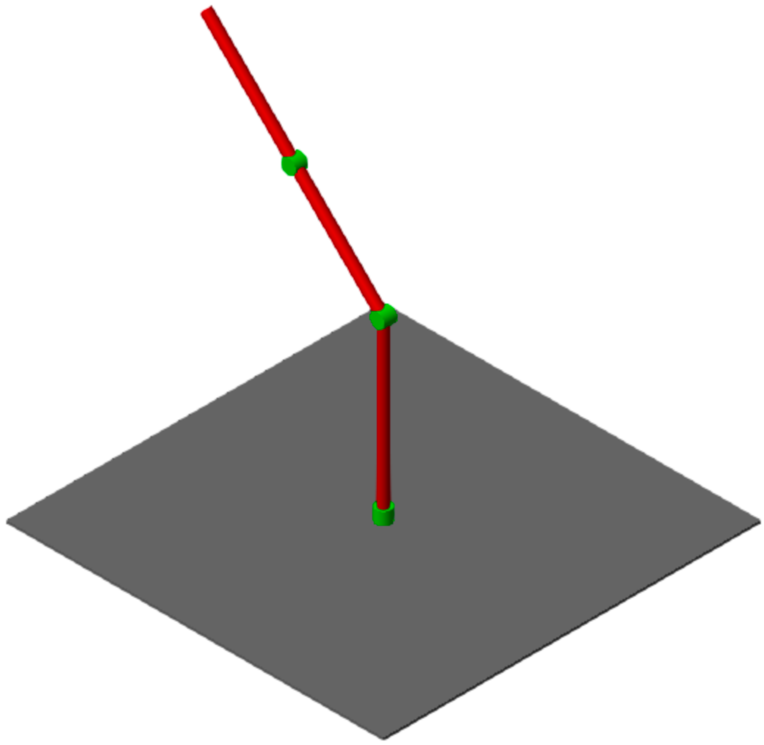
3.2.3 Model Visualization

Using the Simscape model from the inverse kinematics lab, both singularity types were illustrated.

Figure 5: Arm at boundary singularity

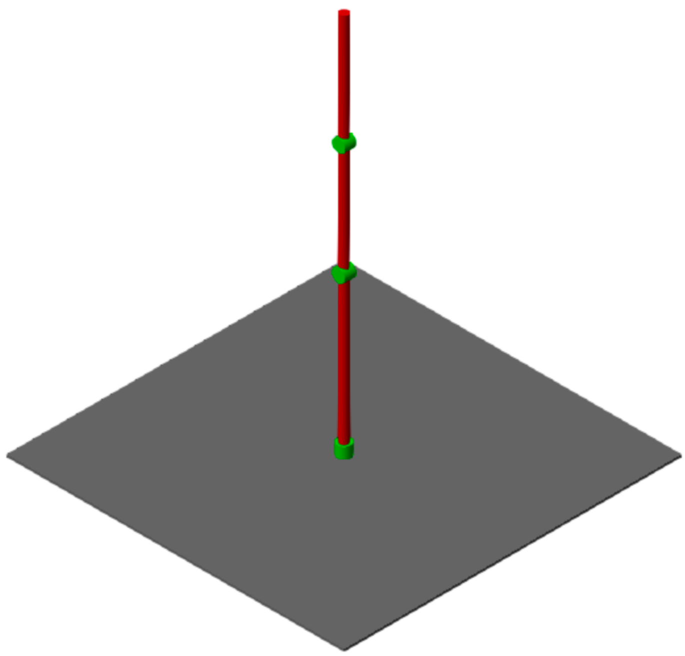


[π, 0, 0]

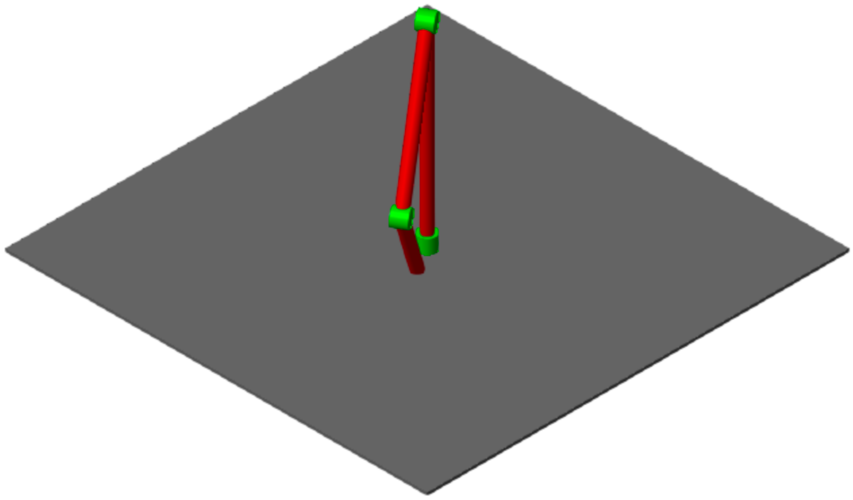


[3π, π/4, 2π]

Figure 6: Arm at shoulder singularity



[0, π/2, 0]



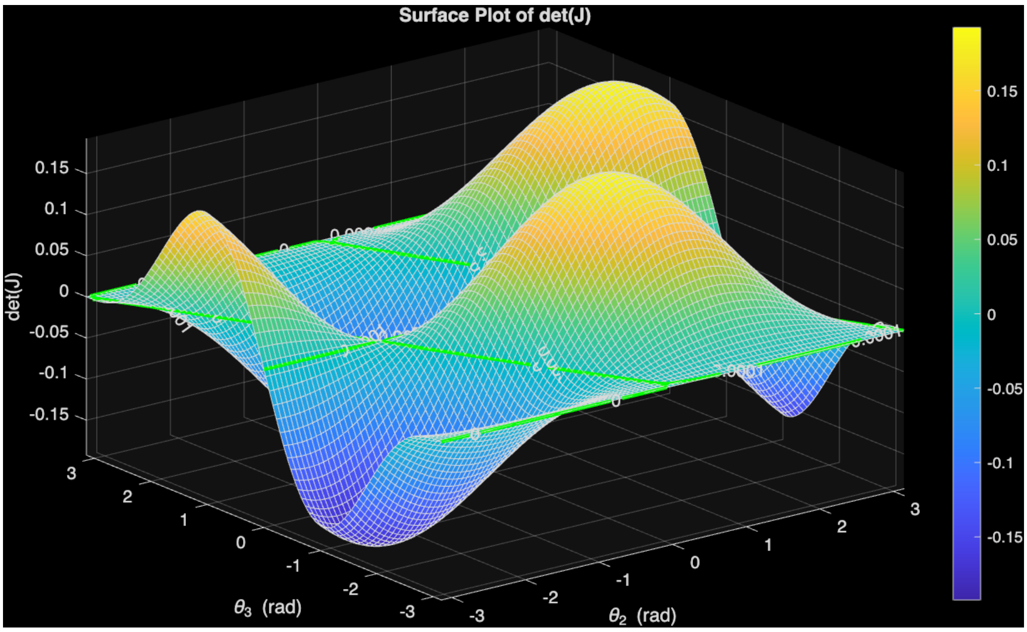
$[-\pi/3, -\pi/3, -\pi/3]$

3.3. Mathematical Analysis of Singularities

3.3.1 Determinant Surface

The determinant of the Jacobian was plotted as a function of joint angles θ2θ2 and θ3θ3.

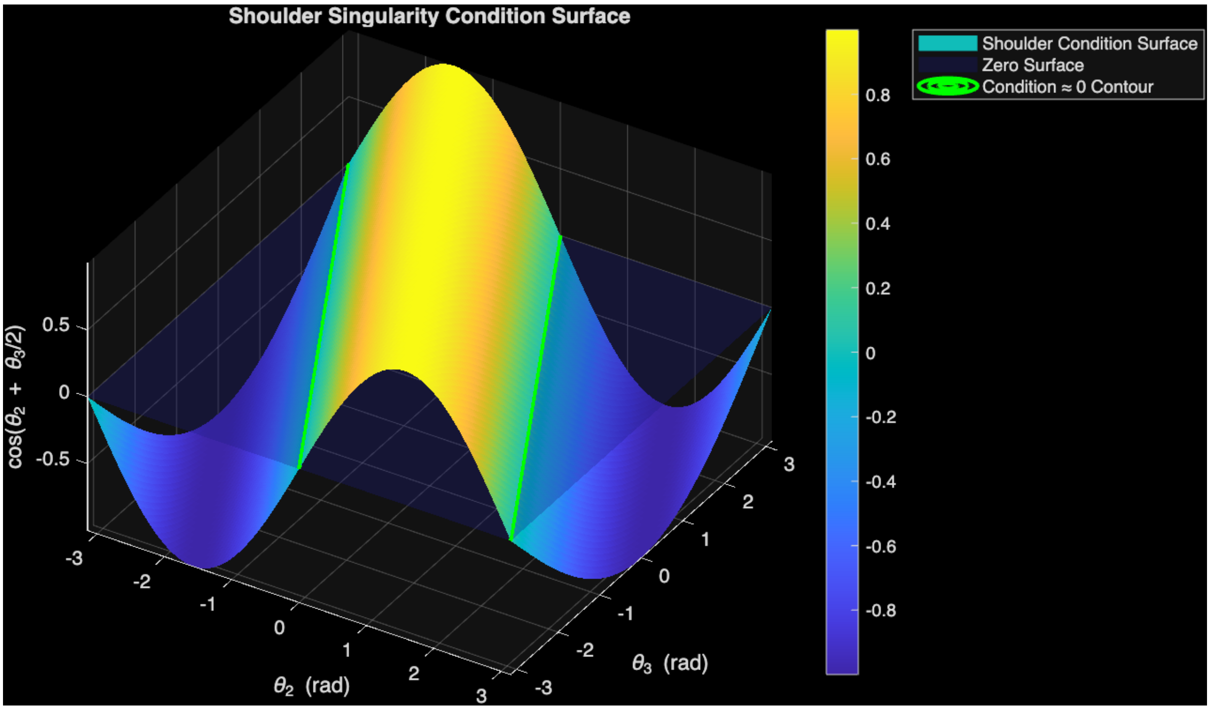
Figure 7: Surface plot of determinant det(J(θ_2, θ_3)) Contour plot showing determinant ≈ 0 region



3.3.2 Shoulder Singularity Condition

The analytical condition for shoulder singularity was derived and plotted.

Figure 9: Surface of shoulder singularity condition vs (θ_2, θ_3) and Contour plot showing shoulder singularity regions



Points where the condition was near zero (<0.01<0.01) were highlighted to identify critical configurations.

3.4. Simscape Path-Following Experiment

3.4.1 Simulation Setup

The Simscape model was configured with:

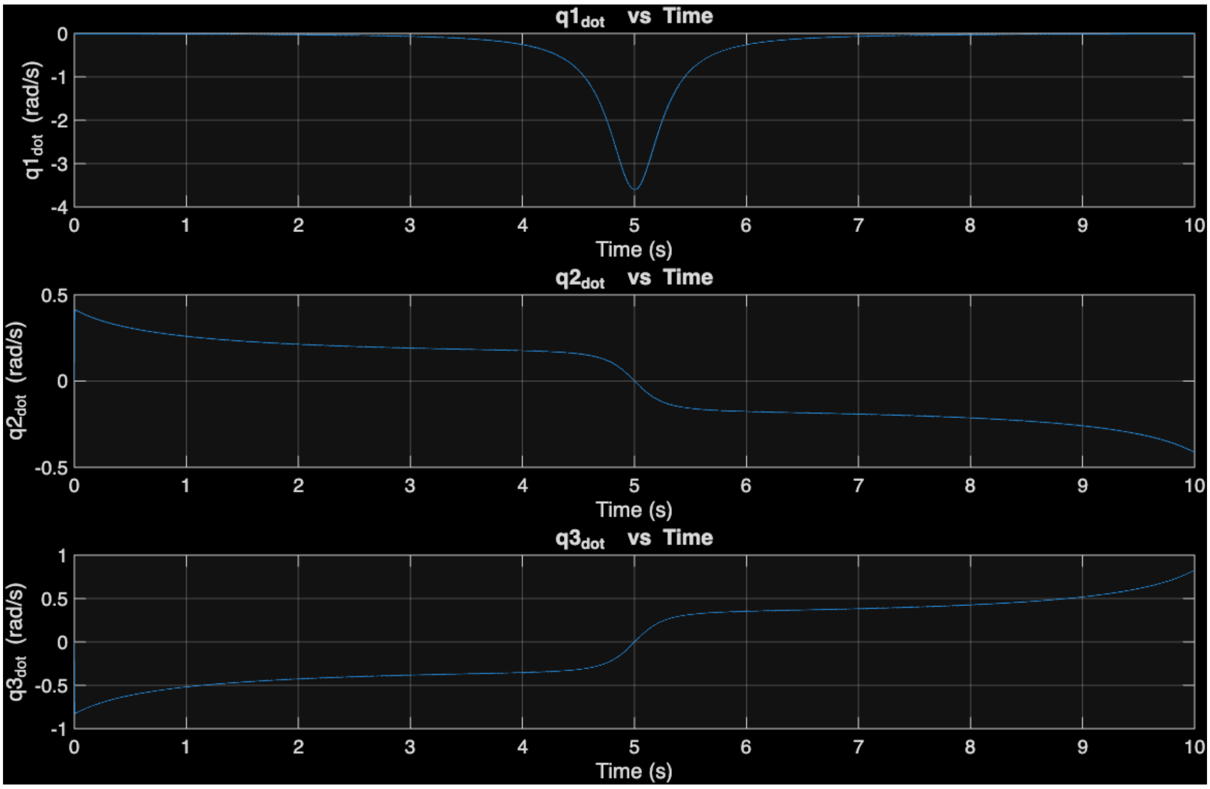
- Path: Straight line along x-axis
- Step size: 0.001 (ode4)
- Simulation time: 10 s
- Initial joint states corresponding to the first point on the path

The path was offset by 0.05 m in the y-direction.

3.4.2 Joint Velocity Behavior

The joint velocities ($\dot{q}_1, \dot{q}_2, \dot{q}_3$) were plotted versus time.

Figure 11: Joint velocities vs. time — approaching singularity



Observation: As the manipulator approached the singular configuration, joint 1 velocity increased sharply, confirming the amplification effect described in the introduction.

3.5. Proximity to Singularity Study

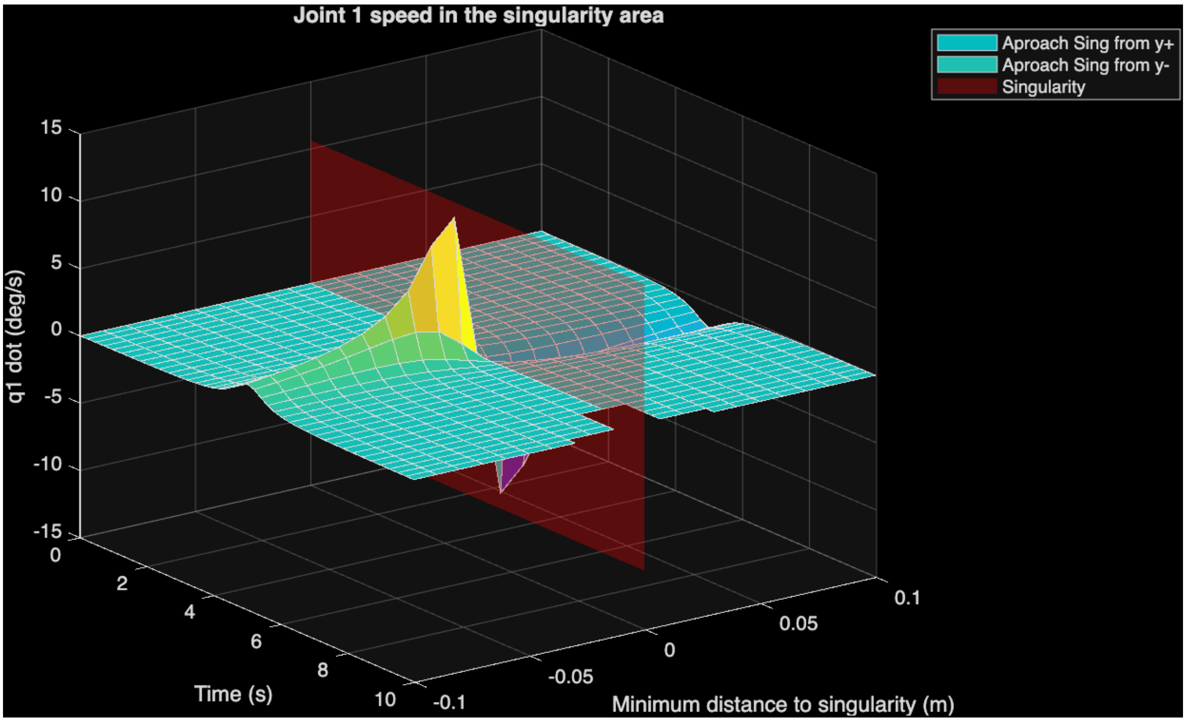
3.5.1 Variable Path Distances

The simulation was repeated for multiple path offsets ranging from -0.1 m to +0.1 m in 21 steps.

3.5.2 Surface Plot of $\dot{q_1}$ vs Distance

A surface plot was generated showing $\dot{q_1}$ as a function of time and minimum distance to the singularity.

Figure 12: Surface plot of $\dot{q_1}$ vs. time and distance to singularity

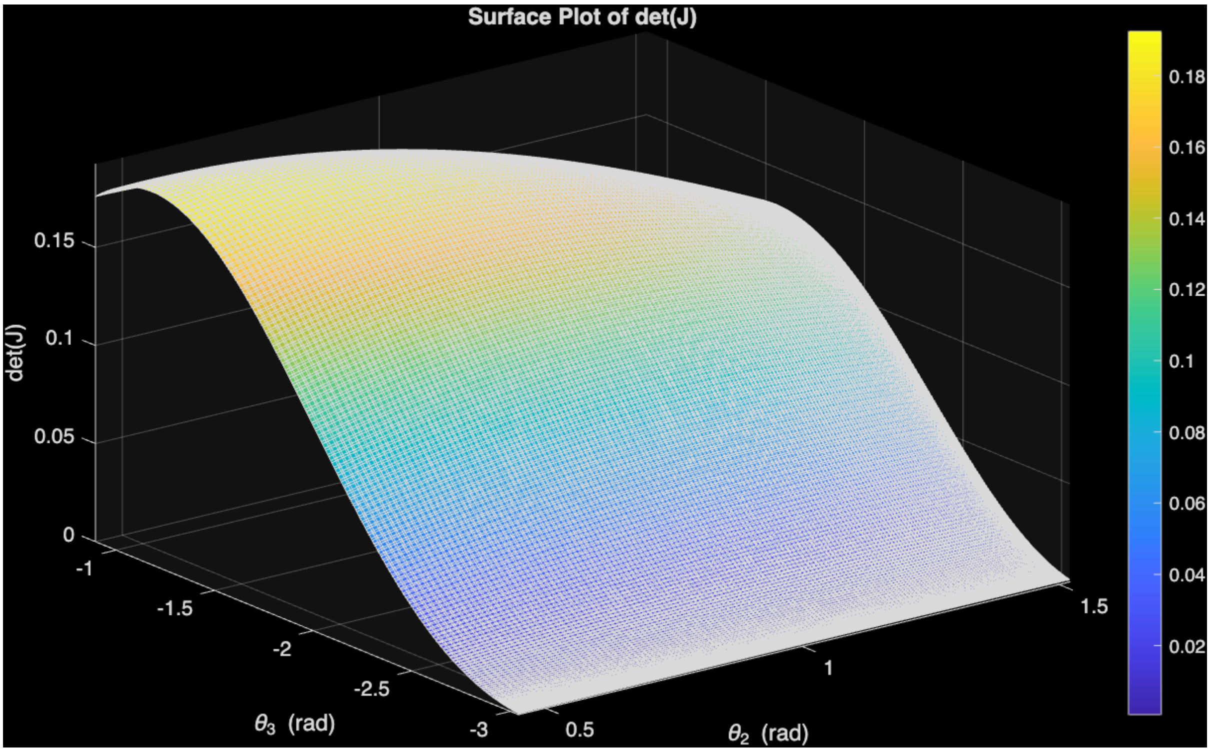


A plane at y = 0 was added to represent the singularity boundary.

3.5.3 Singular Crossing Case

For the case where the path crossed the singularity (y = 0), the corresponding values of q2 and q3q3 were added to the shoulder singularity plot.

Figure 13: q_2 and q_3 overlay on shoulder singularity surface



4. Discussion

This lab demonstrated how the Jacobian determinant predicts both the mathematical and physical behaviour of robotic manipulators. The exercises confirmed the singularity conditions analytically and showed their consequences through simulation.

For the 3-link planar manipulator, the symbolic Jacobian determinant reduced to

$$\det(J)=2\sin(\theta_2)/25$$

This means singularities occur only when $\theta_2=0$, or $\theta_2=\pi$, which correspond to the folded and fully extended positions. These configurations are boundary singularities, marking the edge of the workspace. The plots generated in the analysis matched these conditions exactly.

For the anthropomorphic arm, the determinant depends on multiple joint variables, producing more complex singularity behaviour. Two clear types appeared in the results. Boundary singularities occurred when the arm was fully stretched, similar to the planar model. Shoulder singularities, however, arose when the joint axes aligned, producing an internal singularity that can occur within the reachable workspace. Surface plots of det(J) over θ_2 and $\theta_3$effectively highlighted these regions.

The Simscape simulations demonstrated the practical consequences of operating near a singularity. When the end-effector followed a straight-line path close to a singular configuration, the measured joint velocities showed a pronounced spike in $\dot{q}_1$. This confirmed the expected phenomenon of velocity amplification near singularities. A follow-up experiment quantified this effect: the surface plot of peak joint velocity versus distance from the singularity showed that even small end-effector motions demand disproportionately large joint velocities as the arm approaches a singular point.

Taken together, the findings clearly show that the mathematical condition $\det(J)\rightarrow 0$ directly corresponds to real physical limitations. Boundary and shoulder singularities were both identified and validated through symbolic analysis and simulation. The experiments confirmed that singularities reduce mobility and can cause unbounded joint velocities, underscoring the importance of recognizing and avoiding these configurations in robotic trajectory planning and control.

5. Conclusion

This lab demonstrated the derivation and physical interpretation of the Jacobian in differential kinematics.
By analysing both planar and anthropomorphic manipulators, we confirmed that singularities represent critical configurations where control and motion capabilities are compromised.
Understanding these points is fundamental in robotic control design to ensure safe, stable, and efficient manipulator operation.